Maxwell Fang

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SKILLS

Language Skills: Advanced in English, intermediate in French, basic in Mandarin
Programming Languages: **Python**, **SQL**, **R**, **Matlab**, **Bash, Julia**
Tools: **Git, Docker, Kubernetes, AWS EC2, EKS, ECR, S3, Gstreamer, Tableau, MS SQL Management studio, Gurobi**
Libraries: Pytorch, Numpy, PyGObject, OpenCV, Threading

PROFESSIONAL & LEADERSHIP EXPERIENCE

University of Waterloo Waterloo, Ontario
Continuous Optimization Research Assistant (Volunteer) Oct 2026 – Present

- Tested optimality and runtime of second order cone programs with sparse subspace embedded constraints and obtained 90% decrease in runtime on high-dimensional benchmarks using JuMP and Mosek Optimizer

University of Waterloo Waterloo, Ontario
Undergraduate Research Assistant (Volunteer) Nov 2024 – July 2026

- Developed a method to generate artificial image datasets modeling neural scaling laws using numpy and OpenCV to analyze eigenpairs of the covariance matrix and derived closed-form solutions for eigenvalues of artificial datasets
- Conducted numerical experiments on MNIST and Artificial datasets to understand and determine causes of eigenvector localization in Fourier transformed covariance matrices in artificial datasets

Ford Motor Company of Canada Waterloo, Ontario
Software Developer – AI/ML Jan 2025 – April 2025

- Estimated homography matrices via RANSAC by minimizing reprojection error and applied camera calibration and undistortion to transform Perspective view images to Bird's eye view to validate and visualize lane-line prediction model inferences using **OpenCV** and **numpy**
- Developed image interpolation to remove visual artifacts in transformed Bird's Eye View images and temporal motion smoothing to augment low framerate videos
- Created multi-threaded QA visualization pipeline using **PyGObject Gstreamer** to evaluate accuracy of LIDAR/RADAR ground truths and lane-line prediction models by visualizing inferences on PV/BEV images and converting to smoothed video

Eon Media Toronto, Ontario
Lead AI/ML Engineer Co-op January 2024 – August 2024

- Led and trained a team of 3 coops to maintain pre-existing video processing pipelines, object detection models, NLP models, and develop AI/ML solutions
- Implemented optimization-based unsupervised clustering methods with **Conjugate Gradient Method** and **PCA** to cluster high dimensional image embeddings (2048 dimensions) created by **Pytorch resnet50** and reduce former clustering runtime 75% using **numpy**, **scipy.sparse** and **concurrent.futures**
- Designed and implemented Logo detection model training pipeline to reduce manual labour by 45% and 3rd-party software costs by 80%, featuring image clustering, data cleaning, formatting, and augmentation for **YOLO**-based models
- Optimized and redesigned video processing pipelines using Python **multiprocessing** and **Kubernetes** to decrease CPU usage by 85%, pipeline setup time by 60% and overall processing time by 40%

CAA South Central Ontario Thornhill, Ontario
QA Automation Intern May 2022 - August 2022

Projects

Gesture Controls | *Python, Pytorch, OpenCV, Numpy, Pose Estimation* May 2026 – Present

- Implemented Stacked Parts Based Network with Heatmaps MSE Loss with bone length penalty for 2D Hand Pose Estimation and trained/validated on HanCo and FreiHand datasets with custom dataloaders and data augmentation in **Pytorch**
- Designed multi-threaded video processing pipeline to minimize image buffer streaming with **OpenCV** for running real-time inference on quantized models and clustering hand pose movement embeddings to identify and trigger programmable macros

EDUCATION

University of Waterloo – Faculty of Mathematics Ontario, Canada
Bachelor of Mathematics in Combinatorics and Optimization Sept 2021 - April 2026
Master of Mathematics in Combinatorics and Optimization Incoming Sept 2027